

Harsh Dave

Computational genomics · software

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EDUCATION

Texas A&M University

College Station, TX

M.S. Computer Science

Expected 2028

Graduate coursework includes computer-human interaction and theory of computability.

B.S. Computer Science and B.S. Statistics, Summa Cum Laude

May 2026

GPA 3.92/4.00.

RESEARCH INTERESTS

Computational genomics of *Mycobacterium tuberculosis*. Drug-response experiments, molecular evolution, and software for reading genes and papers.

RESEARCH EXPERIENCE

Graduate Assistant Research, Computational Genomics

Jun 2026 - present

Ioerger Lab, Texas A&M University · Advisor: Thomas R. Ioerger

- Study how gene disruption changes bacterial fitness across drug and growth conditions, using transposon-sequencing data.
- Fit gene-level linear models, compare nested models with likelihood-ratio tests, and test interactions between a drug response and the experimental conditions.
- Apply false-discovery-rate correction across genes and prepare figures and tables for the research team.
- Prepare sequence alignments and run genomic analyses with Python and Slurm on the Texas A&M high-performance computing cluster.
- Compare GenomeMap posterior estimates of dN/dS, the rate of protein-changing substitutions relative to synonymous substitutions.
- Cross-check results with pN/pS mutation counts, multiple-testing correction, and PAML/codeml models.
- Build the lab-facing software described below so gene lookup, paper reading, and study sessions stay next to the analysis.

Undergraduate Researcher

Jan 2025 - May 2025

UrbanResilience.AI Lab, Texas A&M University

- Developed Python data workflows for air-quality and wildfire-response analysis.
- Joined Los Angeles air-quality APIs with scraped wildfire data before the modeling stage.
- Handled differences in format, coverage, timing, and missing values in reusable preparation scripts.

TEACHING

Teaching Assistant, CS 111

Spring 2026

Texas A&M University

- Supported students learning Java and object-oriented programming.
- Reviewed weekly submissions with consistent feedback and explained bugs through questions and small code examples.

ADDITIONAL EXPERIENCE

Software Engineering Capstone, Amazon-sponsored

Aug 2025 - Dec 2025

Texas A&M University · team of three

- Built a breach-intelligence platform connecting automated ingestion, AI-assisted processing, search, graph relationships, and analyst-facing exploration.
- Worked on Scrapy and Tor ingestion through a LangChain and Gemini pipeline for entity and threat processing.
- Designed a data path across AWS S3, MongoDB, Elasticsearch, Redis, and AWS Neptune.
- Delivered Streamlit and Kibana views for querying breach data by sector and geography.

AI Engineering Intern

May 2024 - Aug 2024

Videomagic · remote

- Worked across machine-learning workflow automation, deepfake-detection data, and backend systems.
- Evaluated PyTorch and Hugging Face frame-level workflows for AI-generated video detection, including failure modes.
- Built authenticated APIs with Feathers.js, MySQL, Knex, Autho, and JWT.
- Shipped AI features associated with a 15% lift in engagement.

Cloud Team Research Member

Early undergraduate

SpaceCraft VR · College Station, TX

- Helped an eight-person research team automate cloud infrastructure for simulation work.
- Contributed to a React and TypeScript sandbox and hardened access flows with Autho.

TECHNICAL SKILLS

Data science and statistics. Python, R, SQL, pandas, NumPy, SciPy, scikit-learn, statsmodels, Bayesian inference, hypothesis testing, FDR correction, Matplotlib.

Bioinformatics and computing. GenomeMap, PAML/codeml, dN/dS, pN/pS, FASTA processing, TnSeq, Slurm, Linux.

Machine learning and data systems. PyTorch, Hugging Face Transformers, LangChain, AWS, MongoDB, Elasticsearch, Redis, PostgreSQL, Git, GitHub Actions.

Programming. Java, C++, TypeScript, JavaScript, React, FastAPI, Docker.

Languages. English, Hindi, Gujarati.

SOFTWARE

Independent tools at harsh.bet, built for lab work and for daily use. History for the signed-in tools can sync through a private account. The descriptions below match the public apps.

Research and study

- MtbScope. Search and compare H37Rv genes, with annotations, locations, and links back to sources.
- Research. Import open-access papers by DOI, PMID, or PMCID, and keep text, figures, tables, and references together.
- Quizlet. Flashcards, quiz, fill-in-the-blank, and matching. Decks stay on the device unless sync is turned on.
- Simplify. A study buddy that starts with an analogy and a worked example, then practice, then an explanation in your own words.
- Radar. Papers, Texas A&M events, and paid studies in one list, with duplicates removed and source links kept.

Daily tools

- Today. One view across tasks, habits, nutrition, and training.
- Slate. Lists, due dates, and priorities. Daymark. Habits by count, time, or distance, with streaks and reviews.
- Notes. Folders, labels, search, and optional rich text. Fare. Calories and macros, saved foods, and barcode search.
- Recipes. A searchable library of substantial vegetarian meals with macros and grocery notes.
- Gym. Programs, sets, history, and personal records that stay intact when a program changes.
- ShotLab. On-device basketball shooting-form estimates from video. Timer. A focus timer with drawn scenes such as forest, rain, ocean, and snow.
- Degree. A graduate-plan board that checks catalog rules as courses move between terms.
- Cars. A swipe deck of dealer-listed used cars, with photo galleries, a green flag, a red flag, and a searchable board.

Earlier builds

- PickLedger. Daily sports picks, source records, and graded results.
- Point of sale prototype. Scrum lead for ordering, inventory, analytics, authentication, and accessible customer flows.

COMPETITIONS AND COURSE PROJECTS

Alpha · Tidal Hackathon, 1st place

2024

- A math-learning app with interactive explanations, visualizations, notes, and PDF uploads.
- Input validation reduced tool-call failures from 25% to 6%.

TAMU Datathon, 2nd of 50 teams

2024

- Used web scraping and AI tools to investigate clues during the competition.

Sign Sense

Hackathon

- A sign-language practice app. The team trained a YOLOv5 model on SageMaker and served recognition through FastAPI to a Svelte interface.

ProfFinder • HowdyHack

24-hour build

- Faculty and course search over a database assembled from Texas A&M data, with schedule cards and section search.

Point of sale prototype

Course project

- Scrum lead for a full-stack point-of-sale platform covering ordering, inventory, analytics, authentication, and accessible customer flows.
- Built across React, PostgreSQL, AWS, and OAuth2, with WCAG 2.1 requirements in the customer flows.

COURSEWORK IN PROGRESS

Fall 2026. CSCE 671, Computer-Human Interaction: how people use software, how interfaces are designed, and how to evaluate whether a design works. CSCE 627, Theory of Computability: automata, Turing machines, decidability, and reductions.